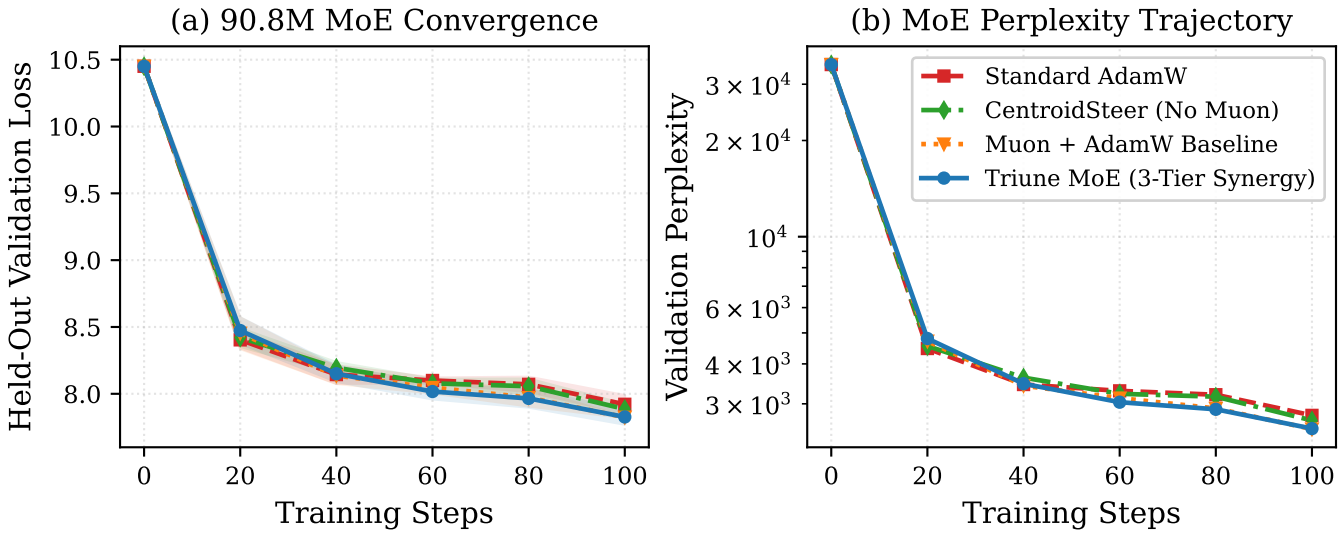
(a) 90.8M MoE Convergence
Held-Out Validation Loss
10.5
10.0
9.5
9.0
8.5
8.0
0
20
40
60
80
100
Training Steps
(b) MoE Perplexity Trajectory
Validation Perplexity
3×10^4
2×10^4
10^4
6×10^3
4×10^3
3×10^3
Standard AdamW
CentroidSteer (No Muon)
Muon + AdamW Baseline
Triune MoE (3-Tier Synergy)
0
20
40
60
80
100
Training Steps